

2024

Time : 3 hours

Full Marks : 35

Pass Marks : 16

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer any five questions.

1. What do you mean by Passive devices ?
How do you find the resistance using color codes ? 2+5 = 7
2. Discuss the Star and Delta connections of capacitors. 7
3. What are N – type and P – type semiconductors ?
Discuss, how the P-N Junction diode works as half-wave rectifier ? 2+5 = 7

4. Draw the circuit diagram of a common emitter NPN and PNP transistors. 7
5. Discuss the role of various components of a CRO. 7
6. Draw the characteristic curves of a P-N junction diode and a Zener-diode. 7
7. Give the basic concepts of Radio Transmittor and Receiver. 7
8. Write a short note on Resonance antenna. 7
9. Write short notes on any two of the following :
 $3\frac{1}{2} \times 2 = 7$

- (a) Lissajous figures
- (b) Transformer
- (c) Yagi Antenna
- (d) MODEM

